

INTRODUCTION

As the part of the B.Tech curriculum under MG University, we, the Sixth Semester Electrical and Electronics Engineering students of Adi Shankara Institute of Engineering & Technology, Kalady have visited K.S.E.B substation ,malayatoor on 7th February 2015 . The IV was successfully completed under the able guidance of Priya miss and Sandeep sir.

This training gave us a rare opportunity to understand the power distribution side and to know about the devices employed for that purpose.

Malayatoor substation is being fed by 110 KV supply from Idamalayar power house to Kalamassery (11MKV feeder) substation through line in line out arrangement .The 110KV feeder from Idamalayar enters this substation as 11MMA feeder and emerges out as 1MARY feeder. Line in line out arrangement minimizes the supply interruption to substation.33KV substation, Kalady is about 10 km from this substation and is under control of 110KV substation, Malaytoor and there are three 5 MVA, 33/11KVA transformers and four 11 KVA outgoing feeders are in service. 33KV substation, Kalady and 33 KV substation Koovapady are being fed by 33KV single circuit with ACSR DOG conductor from 110KV substation Malayatoor.33KV Malayatoor-Kalady substation feeder having 6.348 km length with 87 locations and 33 KV Malayatoor- Koovapady feeder with 4.243 km length with 56 locations.

COMPONENTS OF SUBSTATION

TRANSFORMERS

The power transformers are used to step down the voltage. The transformer core is made of high grade non-ageing cold rolled grain oriented steel laminations. the cooling is provided by means of ONAN (Oil Natural Air Natural) and ONAF (Oil Natural Air Forced). There are fans rotating at very high speed near to fins (used to increase the surface area). Tap changing mechanism is provided for maintaining the secondary voltage constant with varying primary voltage. Tap changing causes change in leakage reactance, core loss, I^2R loss etc. the tap changing means changing the turns ratio. The tapings are provided on the HV side so that it carries minimum current.

Tank is of welded construction. It is stiffened by structured steel. It is provided with 2F earth terminals. HV and LV bushings are transported in crates. Conservator is a cylindrical tank divided into 2 chambers –one for the main tank and other for OLTC. The function of the main conservator is to keep the main tank completely filled with oil. OLTC is fitted with an oil sight window. Explosion vent has a bursting plate and is used as a safety device. When the internal pressure of transformer increases by abnormal phenomenon the diaphragm bursts. In latest models this is replaced by a pressure release valve.

When transformer becomes warm, the oil and gas expands. The gases at the top of the oil expel out and air is drawn into the transformer. This process is called breathing. The air entering is made moisture free by letting it to pass through the breather. It consists of a small container with silica gel crystal. The material is blue when dry and whitish when damp. At the bottom of the container some oil is present along with aluminum silicate in order to absorb dust particles. There are 2 breathers, one for the main conservator tank and the other for OLTC.

Two gauges keep track of the temperature of the oil and winding and an alarm goes off when the temperature rises beyond the limited value. The gauges are fixed on the mounting plate of the transformer. The 2 types of indicators are –Oil Temperature Indicators (OTI)-Winding Temperature Indicators (WTI). There is provision for the control of fan motor by sensing the oil temperature.

The substation has got 5 transformers: two 110/11 kv , two 110/33 kv and one 160 KVA, which is used for station purposes.

- 110/11 kv Tr no 1 : TELK make
10 MVA, YY with OLTC, ONAN Type: SALOCD
KVA = HV:10000/16000
LV:10000/16000
Voltage = HV: 100000
LV :33000
Current = HV:52.5/84.1
LV:175/280
Mass of core in winding :14 tonn
Total mass: 29 tonn
Mass of oil: 7.4 tonn
Max. temperature rise of oil: 45 degree celcius.
- 110/11 kv Tr no 2: Kirloskar make
10 MVA, Delta-Y with OLTC, ONAN
- 110/33 kv Tr no 1: TELK make
16 MVA, YY with OLTC, ONAN/ONAF
- 110/11 kv Tr no 2: TELK make
16 MVA, YY with OLTC, ONAN/ONAF
- 110/11 kv Tr no 2: Kirloskar make
10 MVA, Delta-Y with OLTC, ONAN
- Substation auxiliary TRFR ,KEL make
160 KVA ,delta – Y

ISOLATORS

In order to disconnect a part of system for maintenance and repair isolators are used. It is a knife switch designed to open a circuit under no load. If isolators are to be opened, the CB connected must be opened first. Otherwise there is a possibility of occurrence of a spark at the isolator contacts. After repair, first isolators are closed and then CB. There are two types of isolators- line isolators and bus isolators. For bus isolators, there is no earth switch. During maintenance works the line isolator contacts are opened, so that the three phases trip

simultaneously. For the ease of earthing, dead weights are provided at the end of earthing arms.

CIRCUIT BREAKERS

A circuit breaker is an equipment which can open or close a circuit under normal as well as fault conditions. It can be operated manually from the yard as well as from control room using a pulse trigger under normal condition and automatically under fault condition. For latter relay circuits are used. It consists of a moving and fixed contacts enclosed in strong metal tank. Under normal conditions contacts remain closed and the CT senses full load current. When fault occurs, the over-current in the CT primary winding increases the secondary emf thus energizing the trip coil and moving contacts are pulled down, thus opening the contacts and hence the circuit. The arc produced is quenched by SF6 (outdoor CB) and vacuum (indoor CB).

Ratings : Circuit breaker , AREVA SFS

Type : GLS12

Rated voltage : 145 K,

Normal current :3450A